Mathematical overview

Robust balancing: a selected positive whole-number weight is unsplittable and available once. With the target fixed on the left, each remaining weight goes left, right or off. After any one loss, three distinct weights can guarantee targets 1-3, uniquely with {1,2,3}. Guaranteed coverage of 1-4 is impossible: a two-weight kit covering those four targets must be {1,3}, and the three pairs left by the three possible removals cannot all be {1,3}.

Hidden-mass search: a comparison with a known threshold gives lighter, equal or heavier. Equality identifies at most one distinct candidate. With k comparisons, the largest guaranteed ordered candidate set has 2^(k+1)-1 members: M(0)=1, M(k)=2 M(k-1)+1. Thus two comparisons handle seven candidates, but not eight. This differs from a general three-outcome test with three unrestricted branches.

Equal teams: if all weights form m equal, nonempty groups, the total must be divisible by m; this alone is not sufficient. For 1 through 2r, pairing endpoints gives r teams of total 2r+1. Tray order never creates a new partition.

Children discover three different objectives: surviving a loss, choosing informative comparisons, and partitioning all weights. Grades 2-3 can manipulate and test; grades 4-5 can defend uniqueness and impossibility. Trials show a candidate works in tested cases; complete finite coverage proves that case, while the pair and branch arguments explain the bounds.

Entry routes and readiness

Shared student packet: Grades 2-5. Reading can be adult-supported. P1-3 need reliable sums/differences to 8; P4-5 need ordered candidates and lighter/equal/heavier; P6-7 need group totals up to 9 and conservation of all cards.

Optional K-1 oral entry: P6 with the four small dot-labeled weights, or P1 after reliable totals to 3. The child chooses placements while an adult reads or records. Do not force the whole search tree or unique-kit argument.

Materials and preparation

Per pair: one each of weight cards 1-8, a two-pan mat, up to four small team trays, 24 identical counters, candidate cards 1-8, one opaque envelope or folder, pencils and spare paper. Use dot-and-number labels. Make cards large enough to move and pan areas large enough for the target plus two weights. Remove the lost card physically.

Cards stand for amounts, not actual paper masses. A partner enforces totals or compares a hidden card with a named threshold. A real equal-arm balance is optional and needs calibrated unit bundles. Do not split weights, allow duplicate cards in a selected kit, or use lever distances. Print only the selected starting page; estimated preparation 10-15 minutes with reusable mats. Five sets serve eleven children at three fixed tables, one adult per table; the upper trio rotates roles.

Launch together

Let children handle cards, then show one legal balance from familiar base rules, including a weight beside the target. Remove a weight and let a child check the remaining kit. Before search, enact the printed non-task 8/9/10 example: compare with 9, receive heavier, retain 10. The referee answers from the hidden card and the searcher keeps visible candidates.

First visit and return visits

Choose one investigation for a first 20-35 minute visit after a 2-3 minute launch. P1-3 is the robust-kit route; P4-5 and P6-7 make separate return visits. Keep a pair's kit, comparison plan or partition. These are planning estimates; a productive explanation can occupy the remaining visit.

Problem 1-3: A kit that survives any one loss

Entry, prerequisites and timing

Grades 2-3 start P1 with physical removal; Grades 4-5 continue P2-3 after reliable testing. Allow 20-35 minutes. An adult may record the nine P1 balances while children choose them.

Mathematical facts and solutions

P1: when 1 is removed, target 1 balances by 1+2=3; targets 2 and 3 use their own weights opposite the target. When 2 is removed: 1=1, 2+1=3, 3=3. When 3 is removed: 1=1, 2=2, 3=1+2. Here the first number on each left side is the target; unmentioned weights stay off. The table columns record only added weights, not the target again.

P2: only {1,2,3}, even without the 1-8 cap. For $0<a<b$, positive targets lie among $a,b,b-a,a+b$. To obtain 1, either $a=1$ or $b-a=1$. Covering 2 and 3 then leaves exactly the successful pairs {1,2}, {1,3}, {2,3}. If $a=1$, 2 requires $b=2$ or 3. If $b=a+1$ and $a>1$, obtaining 2 forces $a=2$. Every surviving pair of a robust triple must be one of those three, forcing the triple {1,2,3}. A pooled finite card check suffices for the printed bounded question.

P3: impossible. A pair has at most four distinct positive differences. Covering 1,2,3,4 forces its largest value $a+b=4$, hence the distinct positive pair {1,3}. Three distinct weights cannot leave that same pair after every possible removal. The argument applies to weights of any size.

Hold-back hints

Keep the removed card visibly off the table. Ask whether a remaining weight may join the target. For P2, check a proposed kit against one loss at a time. For P3, ask how many positive amounts two weights can balance.

Depth and further questions

Compare the ordinary optimum {1,3,9} with the robust {1,2,3}: the former cannot balance targets 1 or 2 when 1 is removed because only 3 and 9 remain. Redundancy sacrifices maximum range to protect against failure. Ask whether allowing equal-valued cards changes the conclusion. (It does not: two equal cards a, a balance only a and 2a, so the surviving pairs must still be {1,2}, {1,3} or {2,3}, or {1,3} for targets 1-4.) Do not silently allow them in the printed task.

Facilitator notes

Keep the target fixed and preserve the same proposed kit across all removals. Testing different kits after different losses would not establish robustness.

Problem 4-5: Find a hidden mass with comparisons

Entry, prerequisites and timing

Grades 2-3 maintain candidate cards while a partner supplies exact outcomes. Grades 4-5 explain the bound. Allow 20-30 minutes. No physical mass estimation or guessing is part of the rule.

Mathematical facts and solutions

P4: compare first with 4. Equal means 4. If lighter, compare with 2: lighter means 1, equal means 2, heavier means 3. If heavier than 4, compare with 6: lighter means 5, equal means 6, heavier means 7. Test all seven hidden cards; pairs may share cases and pool a checked plan. A singleton needs no further comparison. The first threshold must be 4 for a guaranteed two-comparison plan, because each remaining side can contain at most three candidates.

P5: no plan handles eight. One last comparison distinguishes at most three distinct candidates. After the first comparison, equality contains at most one candidate and each unequal branch at most three, totaling seven. The outcome labels alone do not justify 3^2=9: their branch sizes have different constraints. Thresholds may be any known whole-number mass, but an equality at a threshold outside the candidate set contributes no candidate.

For k comparisons, apply the same branch argument recursively. A median threshold attains the bound for a consecutive set. The assumption is exact ordered comparison of one hidden mass with a known threshold; no counterfeit-coin or unknown-heavy/light model is present.

Hold-back hints

Lay candidates in order and cover only those contradicted by the answer. Ask what can be learned from equality. For the eight-card bound, imagine how many candidates the last comparison can separate on either side of the first threshold.

Depth and further questions

Three comparisons can distinguish 15 candidates by threshold 8, then 4 or 12, then the middle of the surviving three. Compare a plan minimizing worst-case comparisons with one intended to do well on frequently chosen targets; a probability model would need to be stated.

Facilitator notes

Do not ask the referee to reveal more than the named comparison. One checkable plan plus its surviving candidate branches is enough; no concurrent tally is needed.

Problem 6-7: All weights in equal teams

Entry, prerequisites and timing

K-1 can use P6's first kit orally; Grades 2-3 compare the four kits; Grades 4-5 defend P7 completeness. Allow 15-30 minutes, longer for children inventing their own partition challenges.

Mathematical facts and solutions

P6: 1-4 into two teams: {1,4},{2,3}, each total 5. 1-6 into three: {1,6},{2,5},{3,4}, each 7. {1,2,5} into two is impossible: total 8 requires each group to total 4, but weight 5 cannot fit. 1-8 into four: {1,8},{2,7},{3,6},{4,5}, each 9. These are witnesses, not a claim that every printed successful kit has only one partition.

P7: exactly one partition up to tray order, {1,6},{2,5},{3,4}. Total 21 forces each of the three teams to total 7. Weight 6 must join 1. Then 5 must join 2 because 1 is used. The remaining 3 and 4 form the last team. These forced memberships rule out every alternative, including a proposed three-weight group.

Total divisibility is necessary by conservation of mass, but the {1,2,5} case shows it is not sufficient. Each card is used exactly once and each team is nonempty.

Hold-back hints

Check that every card appears in one tray. For an impossible case, ask what each team total would have to be. For completeness in P7, start with the largest weight rather than list every assignment.

Depth and further questions

Endpoint pairing works for every consecutive kit 1 through 2r. If a number of teams divides r, combine equal pairs in equal-sized collections to obtain that number of equal teams. This gives constructions, not a full classification of all partitions.

Facilitator notes

Keep tray position out of the answer definition. Save one membership picture, not several copies with permuted trays.

Sources, checks and classroom record

Sources and adaptation

Current Week 30, Two-pan weight kits, student packets pp. 1-5 and adult guide pp. 1-4 were inspected. They establish the target-fixed two-pan convention and ordinary coverage/uniqueness questions. This encore changes objectives; the three new arguments are independently derived here.

The base guide cites Bolker, Feuer and Zara, Balance weighing - variations on a theme, for balanced ternary. The paper body was not newly read for these encore tasks, and the robustness/search/partition tasks are not attributed to it.

Laura Givental, Maria Nemirovskaya and Ilya Zakharevich, Math Circle by the Bay, Preface printed pp. viii-x (local PDF pages 9-11): themes with deep context and multiple entry levels, manipulatives, returning to explanations, and no universal estimate for a theme's duration. These are reported principles. The encore tasks, pairing and readiness routes here are local proposals, not classroom results from that book.

Verification

The author recomputed the facts, solutions and extension arguments from the actual tasks and design kernels. Independent mathematical reviews are retained separately with the editable sources.

Mathematical, render and extracted-source rebuild results belong in the release-validation record. No physical fit or procedure pretest and no classroom pilot has occurred.

What to record after a visit

Record each investigation actually used, examples attempted, a saved construction or plan, and where each pair wants to resume. A theme can support several visits; count use by investigation, not by finishing pages.

Record the child's action or words first; write your interpretation separately. Note whether children retained the choices after the launch or needed an adult to operate the investigation. All readiness and timing estimates below are unpiloted hypotheses.

Observe whether the lost weight stays unavailable, whether equality is treated as one candidate, and whether tray order is mistaken for a new partition.

Status

Draft bonus companion. Unpiloted. Digital and mathematical checks do not establish physical fit, procedural rehearsal, staffing success or classroom pacing. Physical pretests have not been performed. Base student pages and base facilitator guide are separate.